Errors

Truncation Error: The error from approximating an infinite sum by a finite sum. Upper Bound of $R_n(x)$.

Round-Off Error: The errors from representations of real numbers to numbers with finite digits and their accumulation due to arithmetic operations.

Round-Off Error of Rational Numbers

$$\frac{1}{3} = 0.333333333333333333\ldots \approx 0.333333 \approx 0.333$$

$$\frac{1}{7} = 0.142857142857142857\ldots \approx 0.142857 \approx 0.143$$

Round-Off Error of Irrational Numbers

$$\pi = 3.141592653589793238\ldots \approx 3.14159265 \approx 3.142$$

$$e = 2.718281828459045235\ldots \approx 2.71828183 \approx 2.718$$

Truncation Error

$$\cos 0.01 \approx P_2(0.01) = 1 - \frac{(0.01)^2}{2!} = 0.999950000000000$$

No Round-Off Error in Decimal Numeral System
Round-Off Error in Binary Numeral System

Truncation Error

$$\cos 0.01 \approx P_4(0.01) = 1 - \frac{(0.01)^2}{2!} + \frac{(0.01)^4}{4!} = 0.999950000416666\ldots$$
$$\approx 0.999950000416667$$

Round-Off Error in Binary and Decimal Numeral System

Errors

Example 01-01: Approximate $f(x) = \cos x$ when $x = 0.01$ using Taylor polynomials with $n = 2, 3$.

$\boxed{n = 2}$

$$\cos 0.01 = P_2(0.01) + R_2(0.01)$$

$$\cos 0.01 \approx P_2(0.01) = 1 - \frac{(0.01)^2}{2!} = 0.999950000000000$$

$$\begin{aligned} R_2(0.01) &= \frac{1}{3!} \cdot \sin \xi \cdot (0.01)^3; \quad 0 < \xi < 0.01 \\ &< \frac{1}{3!} \cdot \sin 0.01 \cdot (0.01)^3 && f(x) = \sin x \text{ is monotonically increasing function when } x \in [0{,}0.01] \\ &< \frac{1}{3!} \cdot (0.01) \cdot (0.01)^3 && \sin x < x \text{ when } x \in \mathbb{R}^+ \\ &= 1.666667 \times 10^{-9} && \text{Upper bound of } R_2(0.01) \end{aligned}$$

$\boxed{n = 3}$

$$\cos 0.01 = P_3(0.01) + R_3(0.01)$$

$$\cos 0.01 \approx P_3(0.01) = 1 - \frac{(0.01)^2}{2!} = 0.999950000000000$$

$$\begin{aligned} R_3(0.01) &= \frac{1}{4!} \cdot \cos \xi \cdot (0.01)^4; \quad 0 < \xi < 0.01 \\ &< \frac{1}{4!} \cdot \cos 0 \cdot (0.01)^4 && f(x) = \cos x \text{ is monotonically decreasing function when } x \in [0{,}0.01] \\ &= \frac{1}{4!} \cdot (1) \cdot (0.01)^4 \\ &= 4.166667 \times 10^{-10} && \text{Upper bound of } R_3(0.01) \end{aligned}$$